

# Bayesian Decision Theory

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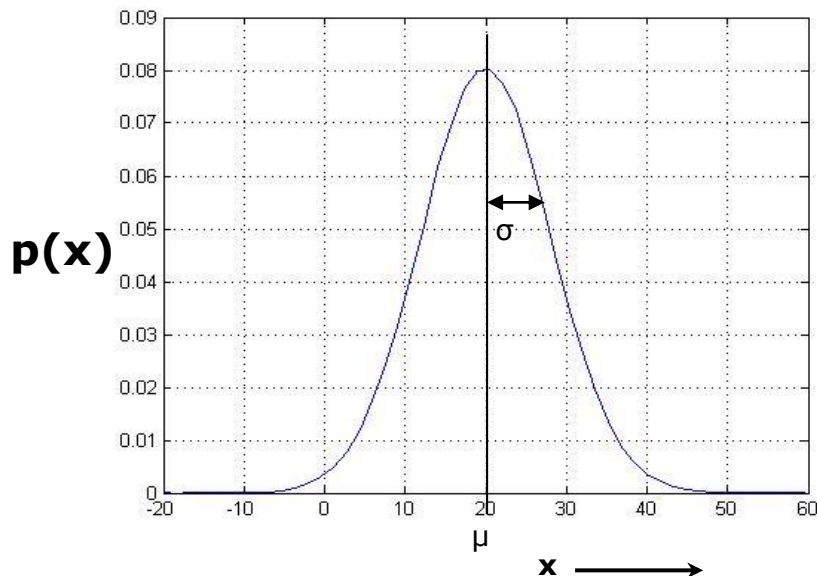
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# Probability Distribution

- Data of a class is represented by a probability distribution
- For a class whose data is considered to be forming a **single cluster**, it can be represented by a **normal or Gaussian distribution**
- **Univariate Gaussian distribution:**

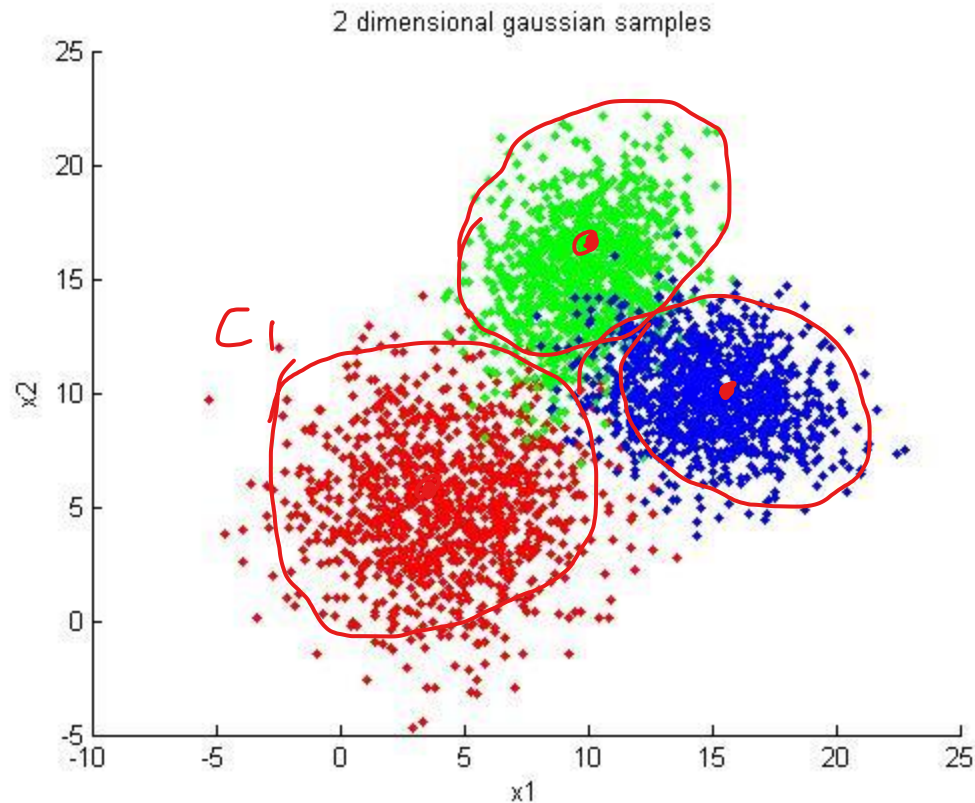
$$p(x) = N(x / \mu, \sigma)$$



$$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- $\mu$  is the mean
- $\sigma^2$  is the variance

# Data Distribution



$6^2$

2D data

- Each class has its own mean and variance

# Gaussian Distribution

Gaussian pdf is extensively used in pattern recognition and machine learning because of

- Mathematical tractability
- Central limit theorem

- ❑ Normal distribution makes math easy-things like calculating moments, correlations between variables, and other calculations that are domain specific.
- ❑ In simple terms, if you have many independent random variables that may be generated by all kinds of distributions, assuming that no thing too crazy happens, the aggregate of those variables will tend toward a normal distribution.

# Multivariate Gaussian Distribution ~~on 4.2~~

- Data in ***d*-dimensional** space

$$p(\mathbf{x}) = N(\mathbf{x} / \mu, C) = \frac{1}{(2\pi)^{d/2} |C|^{1/2}} e^{-\frac{1}{2} (\mathbf{x} - \mu)^T C^{-1} (\mathbf{x} - \mu)}$$

*Handwritten notes:*  $1 \times d$  (for  $\mathbf{x} - \mu$ ),  $1 \times d$  (for  $C^{-1}(\mathbf{x} - \mu)$ ),  $1 \times d$  (for  $\mathbf{x} - \mu$ ),  $d \times d$  (for  $C^{-1}$ ),  $1 \times d$  (for  $\mathbf{x} - \mu$ ),  $d \times 1$  (for  $(\mathbf{x} - \mu)^T$ ).

- $\mathbf{x}$  is the feature vector
- $\mu$  is the mean vector
- $C$  is the covariance matrix

$$d = 2$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} E[x_1] \\ E[x_2] \end{bmatrix}$$

$$C = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$$

*Handwritten notes:*  $2 \times 2$  (for  $C$ ),  $\sigma_{12}$  and  $\sigma_{21}$  are circled.

$$C = \begin{bmatrix} E[(x_1 - \mu_1)^2] & E[(x_1 - \mu_1)(x_2 - \mu_2)] \\ E[(x_2 - \mu_2)(x_1 - \mu_1)] & E[(x_2 - \mu_2)^2] \end{bmatrix}$$

$$x' = [x_1, x_2, \dots, x_d]$$

*Handwritten notes:*  $Cov(x_1, x_2) = \sigma_{12}$ ,  $x_2 =$

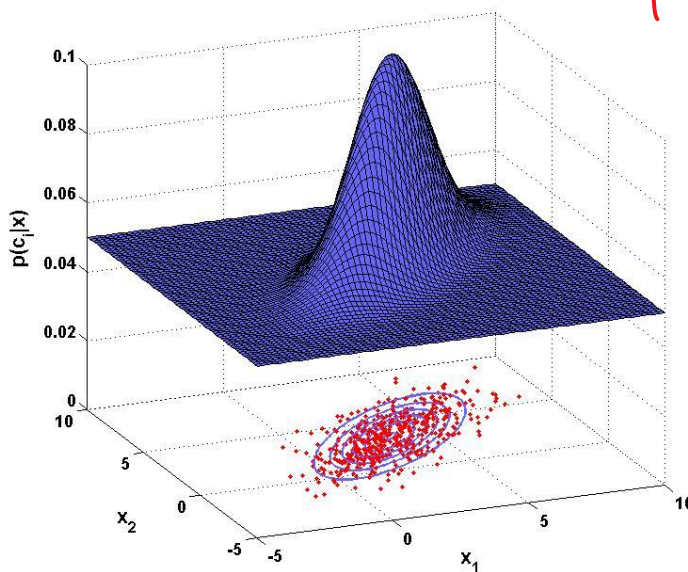
$$x^3 =$$

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} [x_1', x_2', \dots, x_d']$$

*Handwritten notes:*  $x^3 =$ ,  $x_i =$ ,  $\mu =$

# Multivariate Gaussian Distribution

## Bivariate Gaussian distribution:



Handwritten notes in red ink:

$$p(c_1|x) = \frac{p(c_1) p(x|c_1)}{p(x)}$$

> for sample

$$p(c_2|x) = p(c_2) p(x|c_2)$$

$\mu_1$   
 $c_1$

$\mu_2$   
 $c_2$

$x \in c_1$

even  
not  $c_2$

# Covariance formula

Data in  $d$ -dimensional space

- Size of mean vector  $\mu$  :  $\mathbf{1} \times \mathbf{d}$
- Size of  $C$  the covariance matrix:  $\mathbf{d} \times \mathbf{d}$

70:30 iris 150  
3 classes

|           | $f_1$ | $f_2$ | $f_3$ | $f_4$ |
|-----------|-------|-------|-------|-------|
| $x_1$     |       |       |       |       |
| $x_2$     |       |       |       |       |
| $\vdots$  |       |       |       |       |
| $x_{105}$ |       |       |       |       |
| $C_1$     |       |       |       |       |

4x4

$$Cov = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu)^T (x_i - \mu)$$

4x4

$Cov_{f_1, C_1} = \frac{1}{34} \begin{bmatrix} x_1 - \mu & x_2 - \mu & \dots & x_{105} - \mu \end{bmatrix}_{4 \times 1} \begin{bmatrix} x_1 - \mu \\ x_2 - \mu \\ \vdots \\ x_{105} - \mu \end{bmatrix}_{1 \times 4}$

$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix}_{1 \times 4}$

$\frac{1}{34} [(x_1 - \mu)^T (x_1 - \mu) + (x_2 - \mu)^T (x_2 - \mu) + \dots + (x_{105} - \mu)^T (x_{105} - \mu)]$

## Question:

Compute the value of a Gaussian pdf,  $N(\mu, C)$ ,  
at  $x_1 = [0.2, 1.3]^T$  and  $x_2 = [2.2, -1.3]^T$ , where

$$\mu = [0 \ 1]^T \text{ and } C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$C = \Sigma = S$$

## Solution:

Here,  $d = 2$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |C|^{1/2}} e^{-\frac{1}{2} (\mathbf{x} - \mu)^T C^{-1} (\mathbf{x} - \mu)}$$

$$-\frac{1}{2} (\mathbf{x} - \mu)^T C^{-1} (\mathbf{x} - \mu)$$

$$p(\mathbf{x}_1) = 0.1491$$

$$p(\mathbf{x}_2) = 0.001$$



# Bayes Classifier

- Two-class pattern classification using Bayes classifier
- Data of a class is represented by its class-conditional probability density function (likelihood function)
  - Class 1:  $p(\mathbf{x} | c_1)$
  - Class 2:  $p(\mathbf{x} | c_2)$
- The *a priori* knowledge about the occurrence of classes is represented by the **prior** probabilities,  $P(c_1)$  and  $P(c_2)$
- The ***a posteriori*** probabilities are computed using the Bayes rule:

$$P(c_1 | \mathbf{x}) = \frac{p(\mathbf{x} | c_1) P(c_1)}{p(\mathbf{x})}$$

$$P(c_2 | \mathbf{x}) = \frac{p(\mathbf{x} | c_2) P(c_2)}{p(\mathbf{x})}$$

- Probability of occurrence of  $\mathbf{x}$ ,  $p(\mathbf{x})$ , is the normalization factor

## Bayes Classifier (contd.)

- For a test pattern, the class with the higher *a posteriori* probability is assigned using the following Bayes decision rule:

If  $P(c_1 | \mathbf{x}) > P(c_2 | \mathbf{x})$ , assign  $c_1$  to  $\mathbf{x}$ ;  
otherwise assign  $c_2$  to  $\mathbf{x}$

- The above method can be extended for **multi-class pattern classification**.
- Bayes classifier is shown to give the **minimum classification error** when the **probability distributions of classes are known**.

# Bayes Classifier (contd.)

Generalized Bayes decision theory

- $x$  is assigned to  $c_i$

$$P(c_i|x) > P(c_j|x), \quad \forall j \neq i$$

- As  $p(x)$  is positive and the same for all classes

$$p(x|c_i)P(c_i) > p(x|c_j)P(c_j), \quad \forall j \neq i$$

## Question:

Consider a 2-class classification task in the 2-dimensional space, where the data in both classes,  $\omega_1, \omega_2$ , are distributed according to the Gaussian distributions  $N(\mu_1, C_1)$  and  $N(\mu_2, C_2)$

$$\mu_1 = [1, 1]^T, \mu_2 = [3, 3]^T, C_1 = C_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Assuming that  $P(\omega_1) = P(\omega_2) = 1/2$ , classify  $x = [1.8, 1.8]^T$  into  $\omega_1$  or  $\omega_2$ .

## Solution:

$$P(\omega_1 | x) = \frac{p(x | \omega_1) P(\omega_1)}{p(x)}$$

$$P(\omega_2 | x) = \frac{p(x | \omega_2) P(\omega_2)}{p(x)}$$

$$P(\omega_1 | x) = 0.042$$

$$P(\omega_2 | x) = 0.0189$$

x is classified to  $\omega_1$

# Classification: Implementation View

- **Dataset: 100 samples of 4 dimensions (c1: 50, c2: 50)**
  - Training: 70% and Testing 30%
  - Train Set: c1: 35 samples and c2: 35 samples
  - Test Set: c1: 15 and c2: 15 samples

## Training:

- Prior prob of c1 = ?
- Prior prob of c2 = ?
- Mean of c1 = ?
- Cov of c1 = ?
- Mean of c2 = ?
- Cov of c2 = ?

$\mu$   $C_1$   $P(C_1)$   
 $\mu$   $C_2$   $P(C_2)$

## Testing:

- **For each test sample**
  - Calculate the likelihood and posterior probability of each class
  - Assign a class label based on maximum posterior prob

# **Minimum Distance Classifier**

# Euclidean Distance Classifier

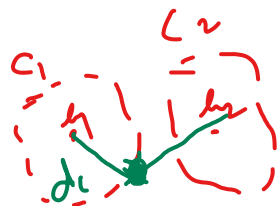
The optimal Bayesian classifier is significantly simplified under the following assumptions:

- ❖ The classes are equiprobable.
- ❖ The data in all classes follow Gaussian distributions.
- ❖ The covariance matrix is the same for all classes.
- ❖ The covariance matrix is diagonal and all elements across the diagonal are equal.

That is,  $C = \sigma^2 I$ ,  $I$  is the identity matrix.

$$C_1 = C_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$x$  is assigned to  $c_i$ , if



$$\sqrt{(x - \mu_i)(x - \mu_i)^T} < \sqrt{(x - \mu_j)(x - \mu_j)^T} \quad \forall i \neq j$$

$d_1 < d_2$

It assigns a pattern to the class whose mean is closest to it w.r.t the Euclidean norm.

$$x \in c_1$$

# Mahalanobis Distance Classifier

If one relaxes the assumptions required by the Euclidean classifier and removes the last one, the Bayesian classifier becomes equivalent to the minimum Mahalanobis distance classifier.

- ❖ The classes are equiprobable.
- ❖ The data in all classes follow Gaussian distributions.
- ❖ The covariance matrix is the same for all classes.

$x$  is assigned to  $c_i$ , if

$$\sqrt{(x - \mu_i)^T S^{-1} (x - \mu_i)} < \sqrt{(x - \mu_j)^T S^{-1} (x - \mu_j)} \quad \forall i \neq j$$

*Handwritten notes:* The first square root term is annotated with "1 x d d x d" and the second with "1 x 1". A bracket is drawn under the first term.



**Thank You**